EXCEL DATA VISUALIZATION

Charts Handbook

Lessons 114-125 | Comparison Charts, Trend Charts and Part-to-Whole Charts

A practical reference for creating, customizing and interpreting charts in Microsoft Excel. Includes copy-ready data tables, exact menu steps, annotated comparisons, use cases and cautions.

Lesson	Topic
114	Why do we need charts?
115	Stacked 2D Column
116	Clustered 2D Column
117	100% Stacked 2D Column
118	Clustered 2D Bar
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120	100% Stacked 2D Bar
121	Line Charts with Markers
122	Area Chart
123	Stacked Area Chart
124	2D Pie Chart
125	Pie of Pie Chart

Based on the sequence and examples in your chart lesson transcript, together with clearly marked illustrative practice datasets.

LESSON 114 | WHY CHARTS MATTER

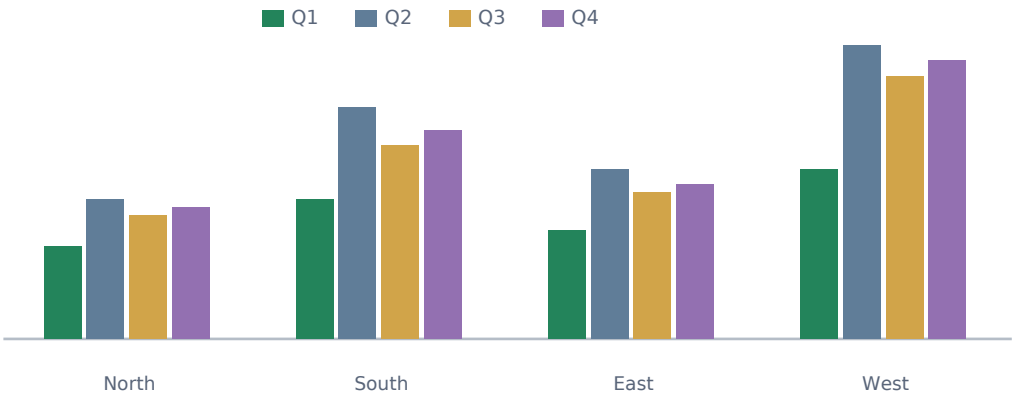
Why Do We Need Charts?

Charts convert summarized business data into visual comparisons. They help management quickly identify high-performing regions, weaker categories and seasonal patterns. They do not replace calculations; they make calculated results easier to interpret.

Practice data: quarterly regional sales

Region	Q1	Q2	Q3	Q4
North	12000	18000	16000	17000
South	18000	30000	25000	27000
East	14000	22000	19000	20000
West	22000	38000	34000	36000

Visual comparison



Steps in Excel

Step	Action
1	Select cells A1:E5.
2	Insert > Insert Column or Bar Chart > Clustered Column.
3	Rename title to Quarterly Sales by Region.
4	Use Data Labels or the value axis to communicate amounts.

Verified practice insights

Region totals: North 63,000; South 100,000; East 75,000; West 130,000. West is highest and South second. Quarter totals: Q1 66,000; Q2 108,000; Q3 94,000; Q4 100,000. Q2 is strongest.

Remember: Choose a chart based on the decision or question: category comparisons, time trends, or composition.

LESSON 115 | STACKED 2D COLUMN

Stacked 2D Column

Places series vertically within each category. Total column height represents the sum, and each section represents a component.

Practice data - paste into Excel

Shop	Small Pizza	Large Pizza
Shop 1	45	55
Shop 2	35	65
Shop 3	60	40
Shop 4	70	30

Create a worksheet named 115_Chart. Paste the table in cell A1. These are illustrative practice values, not exact figures from the video transcript.

Visual example

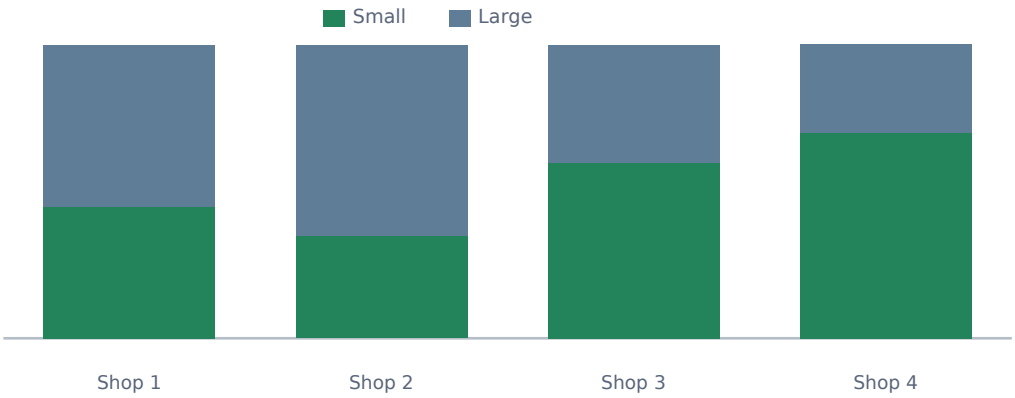


Illustration based on the practice dataset.

How to create it in Excel

Step	Action
1	Select A1:C5.
2	Insert > Insert Column or Bar Chart > 2-D Stacked Column.
3	Rename title to Pizza Sales by Shop.
4	Move legend to Top and enable Data Labels.
5	Remove redundant axes only when numerical context remains clear.

What the chart communicates

• All four shops total 100 in this sample, but their mix differs.

• Shop 4 sells most Small Pizza (70); Shop 2 sells most Large Pizza (65).

• Upper segments can be harder to compare because their baselines move.

When to use it

Use when you need both the category total and its component contributions.

Limitations / when to avoid it

A single series adds little value; too many layers clutter the chart. Use a clustered chart for precise component comparisons.

Important comparison

Chart	Best purpose
Clustered column	Individual categories side by side
Stacked column	Total + components in one vertical column

Quick revision

Stacked 2D Column: Places series vertically within each category. Total column height represents the sum, and each section represents a component.

LESSON 116 | CLUSTERED 2D COLUMN

Clustered 2D Column

Displays two or more series as separate vertical columns sharing a baseline within each category.

Practice data - paste into Excel

Shop	Small Pizza	Large Pizza
Shop 1	45	55
Shop 2	35	65
Shop 3	60	40
Shop 4	70	30

Create a worksheet named 116_Chart. Paste the table in cell A1. These are illustrative practice values, not exact figures from the video transcript.

Visual example

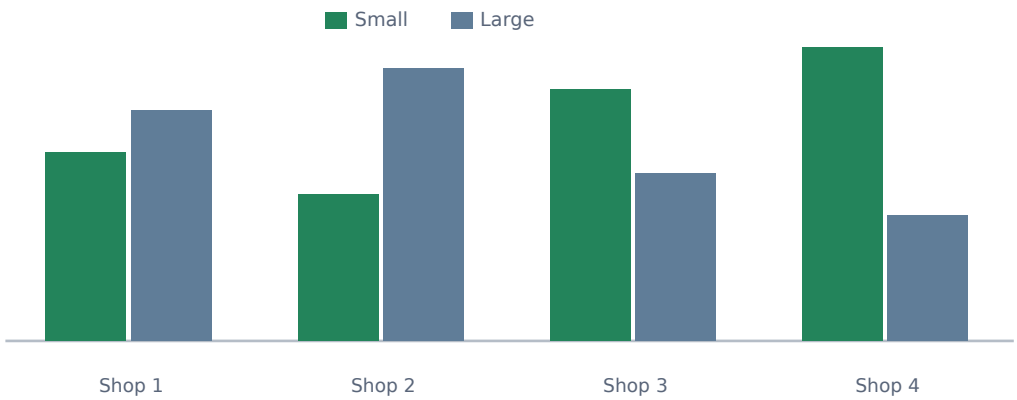


Illustration based on the practice dataset.

How to create it in Excel

Step	Action
1	Select A1:C5.
2	Insert > Insert Column or Bar Chart > 2-D Clustered Column.
3	Rename the title.
4	Enable Data Labels; reposition Legend to Top.
5	Use Chart Design > Switch Row/Column if categories appear incorrectly.

What the chart communicates

- Shop 4 leads Small Pizza sales with 70.
- Shop 2 leads Large Pizza sales with 65.
- Side-by-side common baselines support direct comparisons.

When to use it

Use for comparisons across shops, products, departments, or other categories.

Limitations / when to avoid it

More series in each category produce thinner, more crowded columns. A stacked chart is better to emphasize total composition.

Important comparison

Chart	Best purpose
Clustered column	Direct comparison of individual values
Stacked column	Combined total and component proportions

Quick revision

Clustered 2D Column: Displays two or more series as separate vertical columns sharing a baseline within each category.

LESSON 117 | 100% STACKED 2D COLUMN

100% Stacked 2D Column

Normalizes each category to a column height of 100%. Each segment shows the series share of that category.

Practice data - paste into Excel

Shop	Small Pizza	Large Pizza
Shop 1	45	55
Shop 2	70	130
Shop 3	120	80
Shop 4	140	60

Create a worksheet named 117_Chart. Paste the table in cell A1. These are illustrative practice values, not exact figures from the video transcript.

Visual example

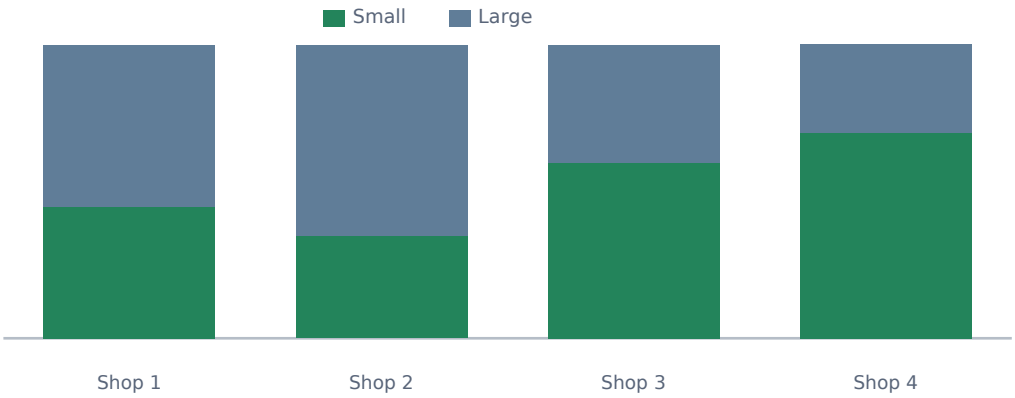


Illustration based on the practice dataset.

How to create it in Excel

Step	Action
1	Select A1:C5.
2	Insert > Insert Column or Bar Chart > 2-D 100% Stacked Column.
3	Rename title to Pizza Sales Composition by Shop.
4	Add Data Labels. For percentage labels, calculate helper percentages and use Format Data Labels > Value From Cells.
5	Move legend to Top and simplify the value axis if appropriate.

What the chart communicates

• Shop 1 is 45% Small and 55% Large.

• Shop 2 is 35% Small and 65% Large.

• Shop 3 is 60% Small and 40% Large; Shop 4 is 70% and 30%.

• The chart does NOT show Shop 2 total (200) as larger than Shop 1 total (100).

When to use it

Use for market share, product mix, or percentage composition across categories.

Limitations / when to avoid it

Avoid for comparing total volume; all columns are the same height. Only one series yields uninformative 100% columns.

Important comparison

Chart	Best purpose
Stacked column	Actual values and category totals
100% stacked column	Relative category composition only

Percentage labels

If Small Pizza is in B and Large Pizza in C, use D2 = B2/SUM($B2:$C2) and E2 = C2/SUM($B2:$C2). Fill down and format as percent.

Default Excel data labels on a 100% stacked chart can display the source values rather than percentages.

Quick revision

100% Stacked 2D Column: Normalizes each category to a column height of 100%. Each segment shows the series share of that category.

LESSON 118 | CLUSTERED 2D BAR

Clustered 2D Bar

Places separate series side by side as horizontal bars within each named category.

Practice data - paste into Excel

Shop	Small Pizza	Large Pizza
Shop 1	45	55
Shop 2	35	65
Shop 3	60	40
Shop 4	70	30

Create a worksheet named 118_Chart. Paste the table in cell A1. These are illustrative practice values, not exact figures from the video transcript.

Visual example

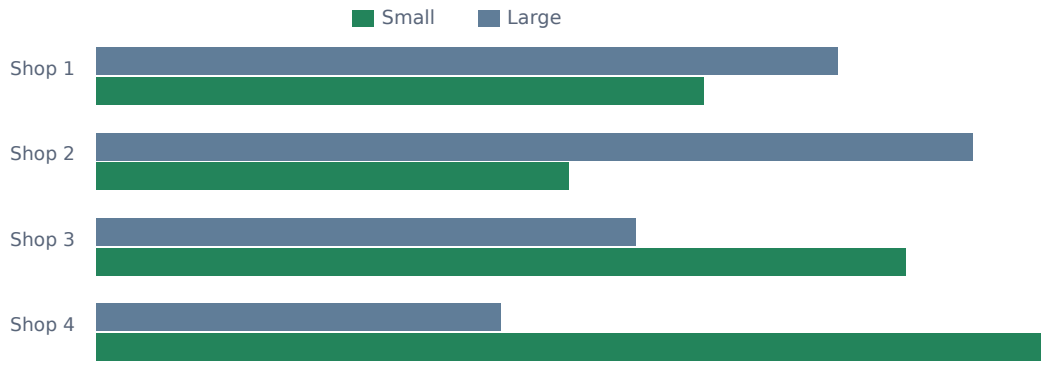


Illustration based on the practice dataset.

How to create it in Excel

Step	Action
1	Select A1:C5.
2	Insert > Insert Column or Bar Chart > 2-D Clustered Bar.
3	Rename the chart.
4	Enable Data Labels, move Legend to Top.
5	Keep category labels; remove the horizontal value-axis labels only if labels make it clear.

What the chart communicates

- Shop 4 has highest Small Pizza sales (70).
- Shop 2 has highest Large Pizza sales (65).
- Horizontal bars help with long category labels.

When to use it

Use for named categories, shops, departments and rankings; particularly effective for longer labels.

Limitations / when to avoid it

Too many series crowd each category. For total-and-contribution comparisons consider Stacked Bar.

Important comparison

Chart	Best purpose
Clustered column	Vertical orientation
Clustered bar	Horizontal orientation for categories

Quick revision

Clustered 2D Bar: Places separate series side by side as horizontal bars within each named category.

LESSON 119 | STACKED 2D BAR

Stacked 2D Bar

Stacks series horizontally. Bar length represents combined category volume, while colored sections show contributions.

Practice data - paste into Excel

Shop	Small	Medium	Large
Shop 1	45	30	55
Shop 2	35	40	65
Shop 3	60	50	40
Shop 4	70	35	30

Create a worksheet named 119_Chart. Paste the table in cell A1. These are illustrative practice values, not exact figures from the video transcript.

Visual example

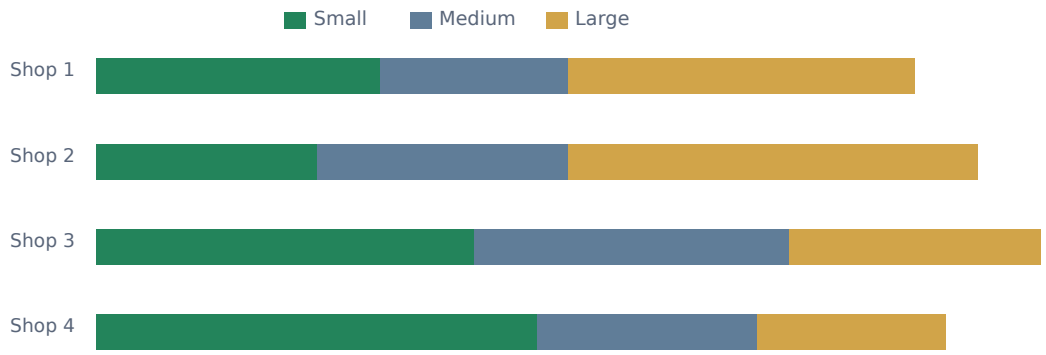


Illustration based on the practice dataset.

How to create it in Excel

Step	Action
1	Select A1:D5.
2	Insert > Insert Column or Bar Chart > 2-D Stacked Bar.
3	Rename title to Pizza Sales by Shop.
4	Add Data Labels > Inside Center and move legend to Top.
5	Retain the shop names on the category axis.

What the chart communicates

• Totals: Shop 1 = 130, Shop 2 = 140, Shop 3 = 150, Shop 4 = 135.

• Shop 3 is highest overall (150).

• Product winners: Small - Shop 4; Medium - Shop 3; Large - Shop 2.

When to use it

Use for comparing category totals and breakdowns with multiple components.

Limitations / when to avoid it

Not ideal for precise comparison of upper stacked segments; overly many series create visual clutter.

Important comparison

Chart	Best purpose
Clustered bar	Individual value comparisons
Stacked bar	Combined totals and component contributions

Quick revision

Stacked 2D Bar: Stacks series horizontally. Bar length represents combined category volume, while colored sections show contributions.

LESSON 120 | 100% STACKED 2D BAR

100% Stacked 2D Bar

Shows each category as a horizontal 100% bar; segments express relative percentages rather than the absolute total.

Practice data - paste into Excel

Shop	Small Pizza	Large Pizza
Shop 1	45	55
Shop 2	70	130
Shop 3	120	80
Shop 4	140	60

Create a worksheet named 120_Chart. Paste the table in cell A1. These are illustrative practice values, not exact figures from the video transcript.

Visual example

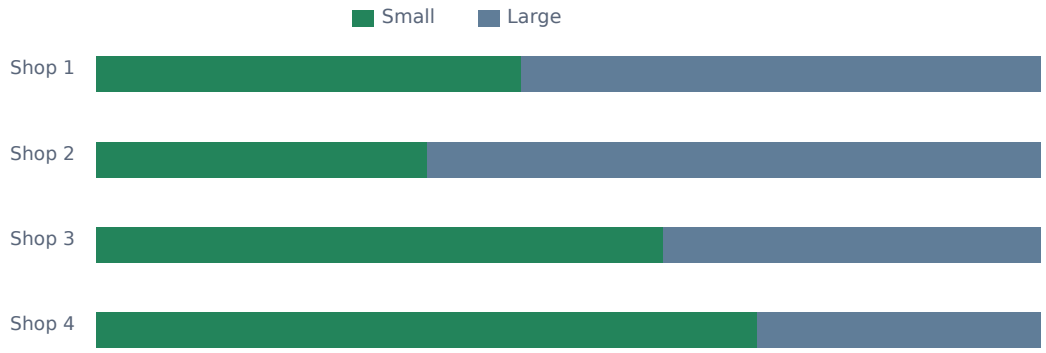


Illustration based on the practice dataset.

How to create it in Excel

Step	Action
1	Select A1:C5.
2	Insert > Insert Column or Bar Chart > 2-D 100% Stacked Bar.
3	Add Data Labels and position Legend on top.
4	For true percentage labels, compute D2=B2/SUM($B2:$C2) and E2=C2/SUM($B2:$C2); format as percentages.
5	In Format Data Labels, use Value From Cells for the correct series.

What the chart communicates

• Shop 1: 45% Small / 55% Large.

• Shop 2: 35% Small / 65% Large.

• Shop 3: 60% Small / 40% Large.

• Shop 4: 70% Small / 30% Large.

When to use it

Use for market shares, budget distribution and product-mix comparison.

Limitations / when to avoid it

It hides differences in raw totals. With a single series, every category becomes a full 100% bar.

Important comparison

Chart	Best purpose
Stacked bar	Compare overall category volume
100% stacked bar	Compare percentage distribution

Quick revision

100% Stacked 2D Bar: Shows each category as a horizontal 100% bar; segments express relative percentages rather than the absolute total.

LESSON 121 | LINE CHARTS WITH MARKERS

Line Charts with Markers

Shows values across ordered time periods using connected lines and markers for individual observations.

Practice data - paste into Excel

Year	Small Pizza	Large Pizza
2021	12000	18000
2022	15000	20000
2023	19000	25000
2024	23000	29000
2025	28000	35000

Create a worksheet named 121_Chart. Paste the table in cell A1. These are illustrative practice values, not exact figures from the video transcript.

Visual example

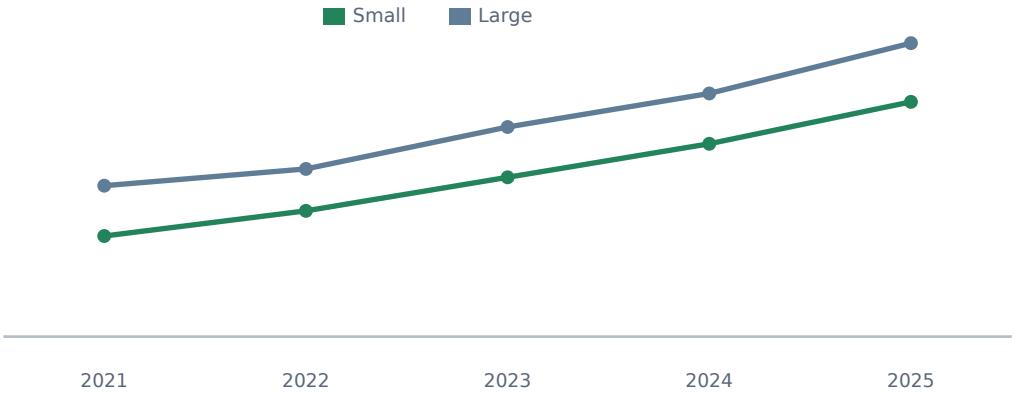


Illustration based on the practice dataset.

How to create it in Excel

Step	Action
1	Select A1:C6.
2	Insert > Insert Line or Area Chart > Line with Markers.
3	Set title to Pizza Sales Trend.
4	For cumulative presentation, use Stacked Line with Markers instead.
5	Move legend to Top and add only useful data labels.

What the chart communicates

• Both products rise from 2021 through 2025.

• Large Pizza exceeds Small Pizza every year.

• Small Pizza rises from 12,000 to 28,000; Large rises from 18,000 to 35,000.

• In a stacked line chart, the upper line is cumulative, not the second series alone.

When to use it

Use for trends over time when specific observations matter. Use stacked line to show a combined trend.

Limitations / when to avoid it

Multiple lines and markers can overlap. Stacked lines with only one series add no analytical value.

Important comparison

Chart	Best purpose
Line with markers	Separate, independent trends
Stacked line with markers	Cumulative contributions over time

Cumulative example

For 2023, Small Pizza = 19,000 and Large Pizza = 25,000; the upper stacked line plots 44,000.

Quick revision

Line Charts with Markers: Shows values across ordered time periods using connected lines and markers for individual observations.

LESSON 122 | 2D AREA CHART

2D Area Chart

Fills the space beneath a trend line, emphasizing both direction and magnitude of numerical values over time.

Practice data - paste into Excel

Year	Large Pizza Sales
2021	18000
2022	20000
2023	25000
2024	29000
2025	35000

Create a worksheet named 122_Chart. Paste the table in cell A1. These are illustrative practice values, not exact figures from the video transcript.

Visual example

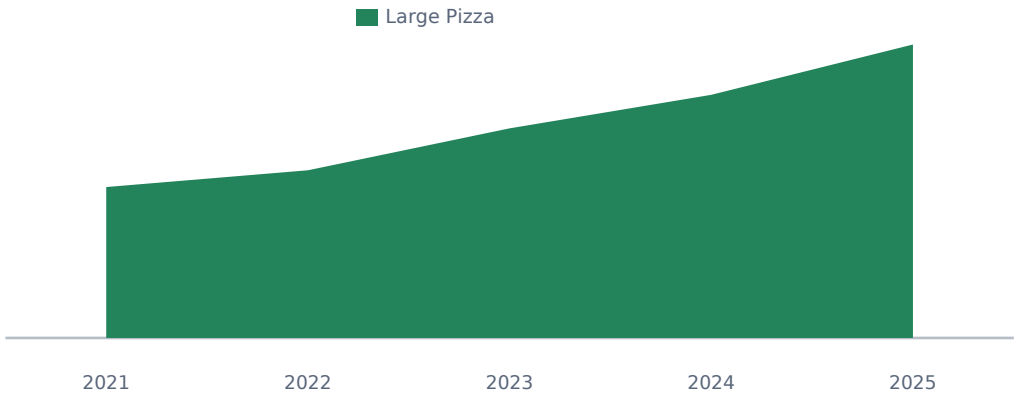


Illustration based on the practice dataset.

How to create it in Excel

Step	Action
1	Select A1:B6.
2	Insert > Insert Line or Area Chart > 2-D Area.
3	Rename title to Large Pizza Sales by Year.
4	Enable selective data labels and simplify the legend for a single series.
5	For multiple overlapping areas, consider adjusting transparency or using Stacked Area.

What the chart communicates

• Sales grow from 18,000 in 2021 to 35,000 in 2025.

• Absolute gain = 17,000; percentage increase is approximately 94.44%.

• Filled area itself does not imply cumulative data.

When to use it

Use for sales volumes, orders, production and other measurements across time.

Limitations / when to avoid it

Overlapping fills can obscure multiple series; labels are easily crowded.

Important comparison

Chart	Best purpose
Line chart	Trend and changes
Area chart	Trend with stronger magnitude emphasis

Quick revision

2D Area Chart: Fills the space beneath a trend line, emphasizing both direction and magnitude of numerical values over time.

LESSON 123 | STACKED AREA CHART

Stacked Area Chart

Stacks filled regions over time, making the combined top boundary and the contribution of each component visible.

Practice data - paste into Excel

Year	Small Pizza	Large Pizza
2021	12000	18000
2022	15000	20000
2023	19000	25000
2024	23000	29000
2025	28000	35000

Create a worksheet named 123_Chart. Paste the table in cell A1. These are illustrative practice values, not exact figures from the video transcript.

Visual example

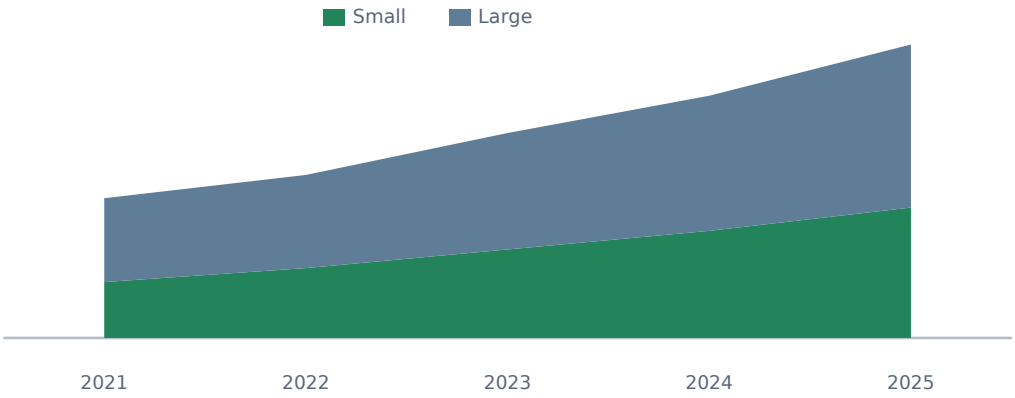


Illustration based on the practice dataset.

How to create it in Excel

Step	Action
1	Select A1:C6.
2	Insert > Insert Line or Area Chart > 2-D Stacked Area.
3	Rename the chart Pizza Sales Contribution by Year.
4	Move Legend to Left or Top.
5	Use clear colors and avoid overcrowding data labels.

What the chart communicates

• Combined sales: 2021 = 30,000; 2022 = 35,000; 2023 = 44,000; 2024 = 52,000; 2025 = 63,000.

• Total grows 110% from 2021 to 2025.

• Upper boundary shows cumulative total; the thickness of each layer shows its series value.

When to use it

Use for total sales or volume over time with contributions of multiple categories.

Limitations / when to avoid it

One series needs no stacking. For precise independent comparisons, line charts may be better.

Important comparison

Chart	Best purpose
Regular area	Separate, potentially overlapping filled areas
Stacked area	Cumulative upper boundary and component thickness

Quick revision

Stacked Area Chart: Stacks filled regions over time, making the combined top boundary and the contribution of each component visible.

LESSON 124 | 2D PIE CHART

2D Pie Chart

Represents a whole as a circle whose slices show each category’s share of the same meaningful total.

Practice data - paste into Excel

Year	Sales
2021	40000
2022	20000
2023	10000
2024	30000

Create a worksheet named 124_Chart. Paste the table in cell A1. These are illustrative practice values, not exact figures from the video transcript.

Visual example

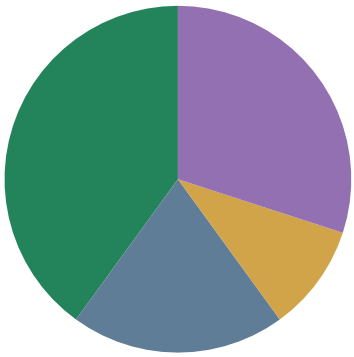


Illustration based on the practice dataset.

How to create it in Excel

Step	Action
1	Select A1:B5.
2	Insert > Insert Pie or Doughnut Chart > 2-D Pie.
3	Rename to Yearly Sales Contribution.
4	Add Data Labels > Format Data Labels > Category Name and Percentage.
5	Remove the legend if slices have clear direct labels.

What the chart communicates

• Grand total is 100,000.

• Year contributions are 40%, 20%, 10% and 30%, respectively.

• 2021 is largest; 2023 is smallest.

• 2021 plus 2024 represent 70% of total sales.

When to use it

Use for small numbers of mutually exclusive categories forming one total, such as market share or budget allocation.

Limitations / when to avoid it

Many slices or similar values are difficult to compare. Do not include a grand total row as an extra slice. Pie charts are not ideal for trends.

Important comparison

Chart	Best purpose
Pie	Part-to-whole with few categories
Bar	Precise category comparisons; many categories

Quick revision

2D Pie Chart: Represents a whole as a circle whose slices show each category’s share of the same meaningful total.

LESSON 125 | PIE OF PIE CHART

Pie of Pie Chart

Uses a primary pie with an Other slice and a second pie that expands selected categories from that Other slice.

Practice data - paste into Excel

Year	Sales
2021	40000
2022	30000
2023	10000
2024	20000

Create a worksheet named 125_Chart. Paste the table in cell A1. These are illustrative practice values, not exact figures from the video transcript.

Visual example

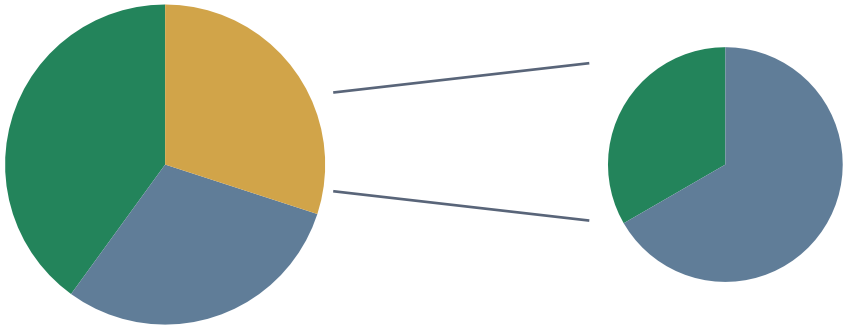


Illustration based on the practice dataset.

How to create it in Excel

Step	Action
1	Select A1:B5.
2	Insert > Insert Pie or Doughnut Chart > Pie of Pie.
3	Right-click a slice > Format Data Series > Series Options.
4	Set Split Series By = Position and Values in Second Plot = 2.
5	Add Category Name and Value labels to the pies, clarifying what Other includes.

What the chart communicates

• Overall sales = 100,000.

• Main pie: 2021 = 40,000; 2022 = 30,000; Other = 30,000.

• Second pie expands Other into 2023 = 10,000 and 2024 = 20,000.

• Shares within second pie are 33.33% / 66.67%; shares of overall total are 10% / 20%.

When to use it

Use when smaller categories deserve a visible, separate breakdown of an existing total.

Limitations / when to avoid it

Secondary pie is not new data; do not double-count it. The built-in Pie of Pie does not automatically drill from annual sales into quarterly records.

Important comparison

Chart	Best purpose
2D Pie	One-level overall distribution
Pie of Pie	Selected categories expanded into a second pie

Splitting options

Split Series By may offer Position, Value, Percentage Value or Custom. Choose Custom when selected categories are not simply the last entries.

Quick revision

Pie of Pie Chart: Uses a primary pie with an Other slice and a second pie that expands selected categories from that Other slice.

QUICK REFERENCE

Choosing the Right Chart

Question	Recommended chart
Compare individual category values	Clustered bar / clustered column
Compare totals plus components	Stacked bar / stacked column
Compare percentages within categories	100% stacked bar / column
Track independent trends across time	Line with markers
Emphasize one volume trend	2D area
Track a combined trend and its components	Stacked area
Show a few parts of one total	2D pie
Expand selected small categories from a pie	Pie of Pie

Excel chart customization checklist

• Write a descriptive title that names the metric and grouping.

• Use appropriate source data ranges; avoid including totals as an extra series.

• Show a legend if series need identification, or label series directly.

• Use data labels only when they make the message easier to read.

• Keep axes if needed to understand scale or maintain accurate comparisons.

• Use consistent colors for the same series across related charts.

• Check if viewers need absolute values, percentages, time trends, or both totals and components.

End of Lessons 114-125.